

Practical No. 4

Aim: To explore the functioning of Platform as a Service and to deploy an application using Heroku / Netlify / Render / Vercel cloud-based development and runtime environment provided by a PaaS platform.

Theory:

Platform as a Service (PaaS) is a cloud computing model that provides a ready-to-use platform for developers to build, deploy, and manage applications without managing the underlying infrastructure such as servers, storage, or networking.

Heroku: Heroku is one of the earliest PaaS platforms supporting multiple languages like Python, Node.js, and Java. However, it discontinued its free tier and is now fully paid, so free alternatives were used in this practical instead.

Netlify: Netlify is a free PaaS platform designed for deploying static websites and frontend frameworks such as React and Vue. It provides GitHub integration, free SSL, and a built-in CDN. Note that Netlify does not support direct deployment of Python or backend applications.

Render: Render is a modern PaaS platform supporting both static sites and dynamic backend services. It offers a free tier with support for Python, Node.js, and Docker, and automatically deploys on every GitHub push.

Vercel: Vercel is a PaaS platform optimized for frontend frameworks, particularly Next.js. It offers a free tier with GitHub-based automatic deployments, serverless function support, and a global CDN.

Advantages of PaaS

- Eliminates infrastructure setup and maintenance overhead
- Faster deployment with continuous integration support
- Free tiers available making it accessible for academic and small-scale projects

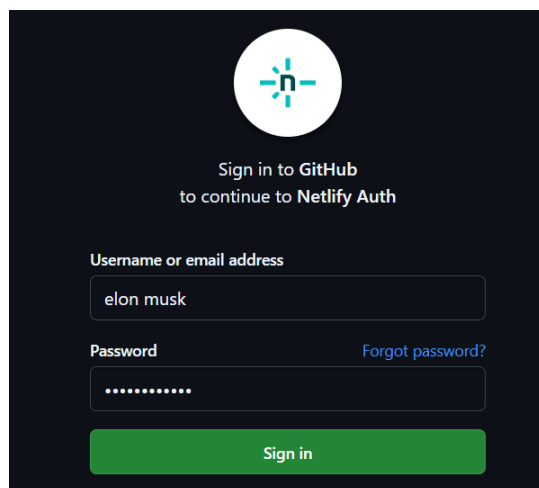
Disadvantages of PaaS

- Limited control over the runtime environment and server configuration
- Free tier has restrictions on bandwidth, build minutes, and uptime
- Vendor lock-in may make migration to other platforms difficult

Steps:

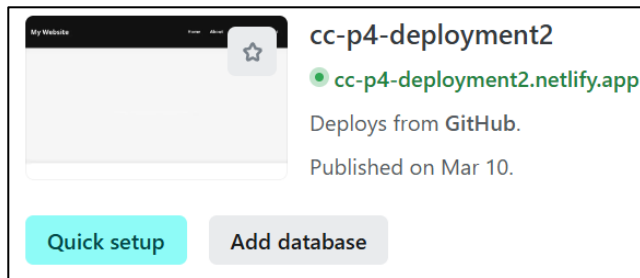
Task 1: Deploy an Application Using Netlify

1. Create a simple HTML application and push it to a GitHub repository.
2. Go to netlify.com and sign up or log in using your GitHub account.



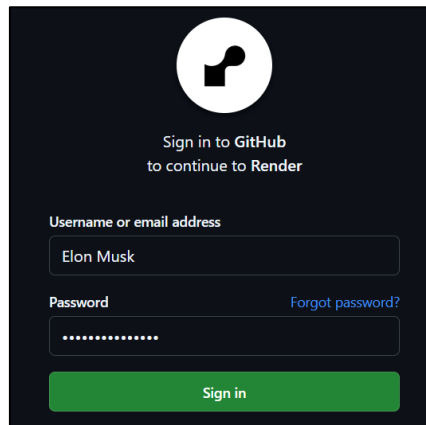
3. Click Add new site and select Import an existing project. Choose GitHub as the source and authorize Netlify to access your repositories.

4. Select the repository containing your application. Leave the build settings as default for a static site and click Deploy site.
5. Once deployment is complete, Netlify provides a live URL. Open the URL in a browser to verify the application is running.

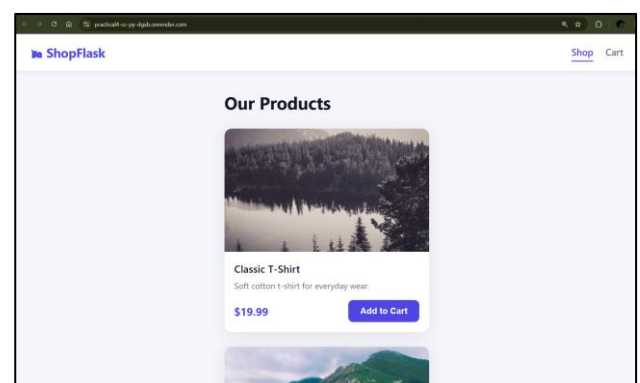
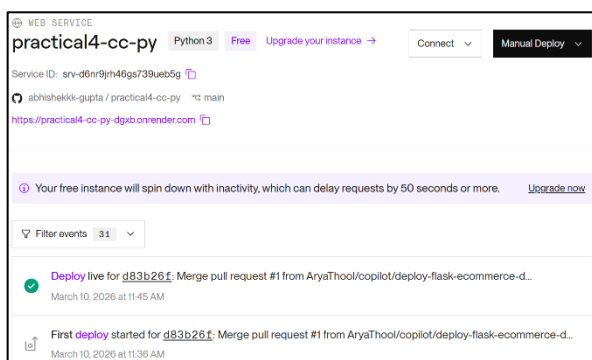


Task 2: Deploy an Application Using Render

1. Go to render.com and sign up or log in using your GitHub account.



2. Click New and select Static Site from the options.
3. Connect your GitHub account and select the repository containing your application.
4. Configure the service name and leave the build and publish settings as default. Click Create Static Site.
5. Once deployed, Render provides a live URL. Open the URL in a browser to verify the application is accessible.



Conclusion: Thus, we have successfully explored the functioning of Platform as a Service and deployed an application using Netlify and Render cloud-based development and runtime environment provided by a PaaS platform.